

Jiaqi LIU

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EDUCATION

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- Oct/24 - Dec/26 Monash University** **VIC, Australia**
- Bachelor of Computer Science, Major in Advanced Computer Science, Faculty of Information Technology
 - **GPA: 75.909 / 100**
- Feb/24 - Oct/24 Monash College** **VIC, Australia**
- Diploma of Information Technology
 - **GPA: 83.25 / 100**
 - Awarded a \$5,000 AUD Merit Scholarship for outstanding academic performance

INTERNSHIP EXPERIENCE

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- Jan/26 - Mar/26 Shopee (SeaBank Digital Bank)** **Shenzhen, China**
- Android Software Engineering Intern*
- Contributed to Android client infrastructure and multi-market engineering optimisation for the Shopee/SeaBank digital banking application across 3 markets, supporting a hybrid Android Native + React Native architecture
 - Fixed inconsistent SHPSSDK protocol usage across 5 modules by enforcing standard-protocol data for the main DFP risk-token exchange and restricting the degraded protocol to fallback paths only, and added atomic (AtomicBoolean) init-state tracking with listener-based gating so RN-side device-info reporting can no longer race SDK initialisation, closing a privacy/risk-control compliance gap
 - Helped build and used a two-tool FastMCP pipeline (compare_apk_files + compare_source_files; Python, decorator-based auto-registration) that let AI clients call jadx-decompilation-based APK diffing and cross-repo Java/Kotlin source diffing as MCP tools, taking a multi-market engineering-merge review from thousands of build artifact files down to about 20 files needing real code review
 - Independently designed and implemented the pipeline's semantic noise-filtering layer (+453/-23 lines across 5 files): normalised 10 categories of decompiler-generated symbol/label renumbering (bytecode labels, obfuscated field/method IDs, Kotlin/R8 synthetic accessors, lambda IDs) to separate real code changes from decompiler noise, cutting flagged diff files by ~40% (500 to 300) and adding structured Manifest/permission/native-library diffing to the report

PROJECT EXPERIENCE

- Aug/26 - present Conversational HEIFA: LLM Agent for Natural-Language Dietary Logging**
- Personal Project | Agent Architecture & Evaluation (TypeScript, Next.js, DeepSeek API)*
- Built a tool-calling LLM agent (TypeScript, Next.js, DeepSeek API) that extracts free-text meal descriptions into structured food-group servings, with a hard architectural separation between semantic extraction (model) and score computation (a deterministic engine), so scores can never be self-reported by the model
 - Enforced code-level source attribution: every field written to the database must carry a verbatim quote from the user's actual message, checked against the real conversation record (not the model's claim) before persistence, eliminating a class of fabrication by construction rather than by prompting
 - Designed and ran a 48-fixture deterministic evaluation harness (no LLM-as-judge) reporting recall, fabrication rate, and paired safety/usability metrics; root-caused and fixed three real regressions to raise recall from 50% to 83.3% while holding fabrication at 0% throughout
 - Ran a controlled ablation showing model self-computed scores carried 34.65 points of arithmetic error alone versus 0 (by construction) for the tool-computed path, then hardened the agent against indirect prompt injection via an explicit untrusted-data boundary, achieving 100% resistance across adversarial fixtures self-designed to target the system's own blind spots
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- Apr/26 - Jul/26 LabLink: High-Concurrency Laboratory Resource Booking Platform**
- Personal Project | Backend Design and Development*
- Designed and implemented a booking platform for laboratory equipment and time slots (resource search, slot management, booking and cancellation, notification, role-based permissions) on Spring Boot, MyBatis-Plus, MySQL, Redis, RabbitMQ and Caffeine, with the core booking path built around stock consistency, asynchronous reliability and observability
 - Performed atomic stock pre-deduction and per-user slot deduplication inside a single Redis Lua script at the request entry, reinforced on the consumer side by Redisson user-slot locks, MySQL conditional

updates and a unique constraint, so concurrent requests can neither oversell inventory nor create duplicate bookings

- Decoupled booking persistence into an asynchronous RabbitMQ pipeline with 3-round delayed retry, dead-letter recording and stock compensation, and built a two-level Caffeine + Redis cache (null-value caching, randomised expiry, mutex rebuild, MQ broadcast invalidation) to handle penetration, breakdown and avalanche
- Implemented booking status notification through a persisted Outbox with MQ event delivery instead of direct in-request calls, serialised 64-bit business IDs as strings to remove front-end JavaScript precision loss, and guarded the entry with a Redis sliding-window rate limiter plus request/failure/retry metrics
- Verified the design through local JMeter load testing: 1,000 resource-read requests sustained about 102 QPS with 0 errors, the two-level cache reached a 99.9% hit rate (1,998 of 2,000 reads), and 50 concurrent booking requests all succeeded with database stock correctly reduced from 500 to 450 and no duplicate records

Mar/26 - Jun/26 Conservation **Deterministic, Stochastic and Game-Theoretic Modelling of Tibetan Antelope**

Modelling and Simulation | Python (NumPy, SciPy, Matplotlib)

- Built a Lotka-Volterra predator-prey model with poaching and an adaptive conservation feedback mechanism, reformulated it as a continuous-time Markov jump process, and ran a Gillespie stochastic simulation over ~4,000 trajectories to estimate extinction probability, minimum viable population size and 95% confidence intervals
- Modelled anti-poaching patrols as a two-player zero-sum game, computing mixed-strategy Nash equilibria via linear programming and feeding the equilibrium poaching pressure back into the stochastic model; validated the framework through stability analysis and Monte Carlo convergence, keeping results reproducible via fixed random seeds

Feb/25 - Oct/25 User Experience Optimisation for Smart Health Management Android Application

Android Development Lead

- Built a multi-screen Kotlin/Jetpack Compose app with an MVVM architecture separating UI from business logic, implementing HEIFA nutrition score parsing/calculation, interactive health data visualisation, Room local persistence and Retrofit networking
- Implemented registration, authentication and multi-user account management with a separate clinician dashboard role, delivering an end-to-end workflow from user data collection and score calculation to DeepSeek-API-generated personalised recommendations

MISCELLANEOUS INFORMATION

Languages:

- Advanced proficiency in Java / Python / Kotlin / SQL / JavaScript / TypeScript
- Familiar with Haskell / Lua / HTML+CSS / LaTeX

Frameworks & Backend:

- Proficient in Spring Boot / MyBatis-Plus / Redis (Lua, Redisson) / RabbitMQ / Caffeine / Next.js / MCP (FastMCP) / OpenAI-compatible LLM APIs
- Experienced in high-concurrency backend design: distributed locking, asynchronous messaging and retry, multi-level caching, rate limiting, REST API design

Deep Learning:

- Feedforward NNs, CNNs, Vision Transformers, RNNs/LSTMs, Transformers, GANs, Diffusion Models; backpropagation, optimisation and LoRA/Adapter fine-tuning

Algorithms & Data Structures:

- String algorithms (KMP, Boyer-Moore, suffix trees/arrays, Burrows-Wheeler Transform) and data compression (Huffman, LZ77)
- Graph & combinatorial optimisation (network flow, bipartite matching, linear programming, DP/greedy) and advanced data structures (B-trees, Fibonacci heaps) with amortised complexity analysis

Databases & Data Tools:

- MySQL / MongoDB / DataGrip / Tableau / Vega-Lite

Development Tools:

- IntelliJ IDEA / VS Code / Android Studio / PyCharm / Git / JMeter / Postman / Overleaf